

3.13.3 Analogue signal processing (A-level only)

3.13.3.1 LC resonance filters (A-level only)

Content

Resonant frequency, $f_0 = \frac{1}{2\pi\sqrt{LC}}$

Only parallel resonance arrangements are required.

Analogy between LC circuit and mass–spring system.

Inductance as mass analogy.

Capacitance as spring analogy.

Derivation of the equation is not required.

Energy (voltage) response curve.

The response curve for current is not required.

Q factor, $Q = \frac{f_0}{f_B}$

f_B is the bandwidth of the filter at the 50% energy points.

3.13.3.2 The ideal operational amplifier (A-level only)

Content

Operation and characteristics of an ideal operational amplifier:

- power supply and signal connections
- infinite open-loop gain
- infinite input resistance.

Open-loop transfer function for a real operational amplifier, $V_{\text{out}} = A_{\text{OL}}(V_+ - V_-)$

Use as a comparator.

The operational amplifier should be treated as an important system building block.